

Osteoporosis, aging, tissue renewal, and product science

Osteoporosis, aging, tissue renewal, and product science The incidence of osteoporosis, like obesity, has been increasing in recent decades. The number of hip fractures in many countries has doubled in the last 30 or 40 years (Bergstrom, et al., 2009). An exception to that trend was Australia in the period between 2001 and 2006, where the annual incidence of hip fractures in women over 60 years old decreased by 28.3%. During those years, the number of prescriptions for "hormone replacement therapy" decreased by 54.6%, and the number of prescriptions for bisphosphonate increased by 245%. The publication of the Women's Health Initiative results in 2002 (showing that the Prem-Pro treatment caused breast cancer, heart attacks, and dementia), led to a great decrease in the use of estrogen treatments everywhere. After the FDA approved estrogen's use in 1972 for the prevention of osteoporosis the number of women using it increased greatly, and by 1994, 44% of women in the US had used it. After the WHI results were published, the number of prescriptions for "HRT" fell by more than half, and following that decrease in estrogen sales, the incidence of breast cancer decreased by 9% in women between the ages of 50 and 54. With the incidence of hip fractures increasing while the percentage of women using estrogen was increasing, it seems likely that there is something wrong with the theory that osteoporosis is caused by an estrogen deficiency. That theory was derived from the theory that menopause was the consequence of ovarian failure, resulting from the failure to ovulate and produce estrogen when the supply of eggs was depleted. The theory was never more than an ideological preference, but the estrogen industry saw it as an opportunity to create a huge market. There are many studies that seem to imply that the greater incidence of osteoporotic fractures among women is the result of their exposure to estrogen during their reproductive years. This would be analogous to the understanding that it is the cumulative exposure to estrogen that ages the nerves in the hypothalamus that control the cyclic release of the gonadotropic hormones, causing the menopause. . . . the nature of science itself changed around the middle of the last century, becoming product and disease oriented, so that now relatively few people are continuing to study bones objectively. Animal studies show that estrogen stunts growth, including bone growth. The high estrogen levels in girls' teen years and early twenties accounts for the fact that women's bones are lighter than men's. In rat studies, treatment with estrogen was found to enlarge the space between the jawbone and the teeth, which is a factor in periodontal disease (Elzay, 1964). Teeth are very similar to bones, so it's interesting that treating male or female rats with estrogen increases their incidence of tooth decay, and removing their gonads was found to decrease the incidence (Muhler and Shafer, 1952). Supplementing them with thyroid hormone decreased the incidence of cavities in both males and females (Bixler, et al., 1957). One of the "estrogen receptors" appears to actively contribute to bone loss (Windahl, et al., 1999, 2001). Studies in dogs following the removal of their ovaries found that there was an increase of bone remodeling and bone formation rate in the first month, followed by a few months of slowed bone formation, but that by 10 months after the surgery the bones had returned to their normal remodeling rate, and that "at no time was a significant reduction in bone volume detected" (Boyce, et al., 1990). With the removal of the ovaries, the production of progesterone as well as estrogen is affected, but the adrenal glands and other tissues can produce those hormones. Until the influence of the estrogen industry overwhelmed it, ordinary science was studying bone development in comprehensive ways, understanding its biological roles and the influences of the environment on it. But the nature of science itself changed around the middle of the last century, becoming product and disease oriented, so that now relatively few people are continuing to study bones objectively. The outstanding physical-chemical property of bone is that it is a reservoir-buffer of carbon dioxide, able to bind huge amounts of the gas into its structure. When carbon dioxide increases in the bloodstream it is at first absorbed rapidly by the bones, and if the blood level of CO₂ is kept high day after day, the rate of absorption gradually slows down, but in experiments that have continued for several weeks the bones were still slowly absorbing more carbon dioxide; the absorption curve seems to be asymptotic. When people move to or from high altitudes, their bones appear to

continue adapting to the different gas pressures for years. A reduction of atmospheric pressure (which allows the tissues to retain more carbon dioxide) helps to reduce the calcium loss caused by immobility (Litovka and Berezovs'ka, 2003; Berezovs'kyi, et al., 2000), and promotes the healing of damaged bone (Bouletreau, et al., 2002). Ultrasound treatment, which accelerates bone healing, stimulates processes similar to reduced oxygen supply (Tang, et al., 2007). The mineral in newly formed bone is calcium carbonate, and this is gradually changed to include a large amount of calcium phosphate. Besides forming part of the mineral, carbon dioxide is also incorporated into a protein (in a process requiring vitamin K), in a process that causes this protein, osteocalcin, to bind calcium. The osteocalcin protein is firmly bonded to a collagen molecule. Collagen forms about 30% of the mass of bone; several percent of the bone consists of other organic molecules, including osteocalcin, and the rest of the mass of the bone consists of mineral. Thyroid hormone is essential for forming carbon dioxide. In the early 1940s, experimental rabbits were fed their standard diet, with the addition of 1% desiccated thyroid gland, which would be equivalent to about 150 grains of Armour thyroid for a person. They became extremely hypermetabolic, and couldn't eat enough to meet their nutritional needs for growth and tissue maintenance. When they died, all of their tissues weighed much less than those of animals that hadn't received the toxic dose of thyroid, except for their bones, which were larger than normal. Experiments with the thin skull bones of mice have shown that the active thyroid hormone, T₃, increases the formation of bone. To increase cellular respiration and carbon dioxide production, T₃ increases the activity of the enzyme cytochrome oxidase, which uses copper as a co-factor. Increased thyroid activity increases the absorption of copper from foods. There is an inherited condition in humans, called osteopetrosis or marble bone disease, caused by lack of a carbonic anhydrase enzyme, which causes them to retain a very high level of carbon dioxide in their tissues. Using a chemical that inhibits carbonic anhydrase, such as the diuretic acetazolamide, a similar condition can be produced in animals. Acetazolamide inhibits the bone resorbing actions of parathyroid hormone, including lactic acid formation and the release of the lysosomal enzyme, beta-glucuronidase (Hall and Kenny, 1987). While lactic acidosis causes bone loss, acidosis caused by increased carbonic acid doesn't; low bicarbonate in the body fluids seems to remove carbonate from the bone (Bushinsky, et al., 1993), and also mineral phosphates (Bushinsky, et al., 2003). The parathyroid hormone, which removes calcium from bone, causes lactic acid to be formed by bone cells (Nijweide, et al., 1981; Lefebvre, et al., 1986). Lactic acid produced by intense exercise causes calcium loss from bone (Ashizawa, et al., 1997), and sodium bicarbonate increases calcium retention by bone. Vitamin K₂ (Yamaguchi, et al., 2003) blocks the removal of calcium from bone caused by parathyroid hormone and prostaglandin E₂, by completely blocking their stimulation of lactic acid production by bone tissues. Aspirin, which, like vitamin K, supports cell respiration and inhibits lactic acid formation, also favors bone calcification. Vitamin K₂ stimulates the formation of two important bone proteins, osteocalcin and osteonectin (Bunyaratavej, et al., 2009), and reduces the activity of estrogen by oxidizing estradiol (Otsuka, et al., 2005). The formation of eggshell, which is mostly calcium carbonate, is analogous to the early stage of bone formation. In hot weather, when chickens pant and lower their carbon dioxide, they form thin shells. A sodium bicarbonate supplement improves the quality of the eggshell (Balnave and Muheereza, 1997; Makled and Charles, 1987). Chickens that habitually lay eggs with thinner shells have lower blood bicarbonate than those that lay thick shelled eggs (Wideman and Buss, 1985). One of the arguments for stopping the sale of DDT in the US was that it was threatening to cause extinction of various species of bird because it caused them to lay eggs with very weak shells. Several other synthetic estrogenic substances, ethynylestradiol, lindane, PCBs, cause eggshell thinning, partly by altering carbonic anhydrase activity (Holm, et al., 2006). Estrogen and serotonin activate carbonic anhydrase in some tissues, progesterone tends to inhibit it. DDE, a metabolite of DDT, reduces medullary bone formation in birds (Oestricher, et al., 1971) and bone mineral density in men (Glynn, et al., 2000). Among its estrogenic effects, DDE increases prolactin (Watson, et al., 2007); one form of DDT inhibits progesterone synthesis and increases estrogen (Wojtowics, et al., 2007). In youth, the mineralization of the collagen framework is slightly lower than in maturity, and the bones are more flexible. With aging, the mineralization increases progressively, and the proportion of collagen decreases slightly, and the bones become increasingly brittle. (Rogers, et al., 1952; Mbuyi-Muamba, et al., 1987). Collagen is a major part of the extracellular substance everywhere in the body, and its concentration increases with aging in the non-calcified tissues. There is considerable renewal and modification of collagen, as new molecules are formed and old molecules broken down, but its average structure changes with aging, becomes less soluble and more rigid,

as the result of chemical cross-links formed between molecules. These cross-links are involved in regulating the differentiation of bone cells (Turecek, et al., 2008). Recently (August 2, 2011), Deasey et al., have published evidence showing that cross-linking is required for bone mineralization (2011). The outstanding physical-chemical property of bone is that it is a reservoir-buffer of carbon dioxide, able to bind huge amounts of the gas into its structure. Around 1950, Fritz Verzar began studying the changes of collagen that occur with aging, and his work led to the "collagen theory of aging." He showed that older, stiffer, less elastic tendons have a higher "melting" or contracting temperature than young tendons. (This effect is responsible for the curling of a piece of meat when it is frying.) Verzar and his colleagues investigated the effects of hormonal treatments on the aging of rat collagen, especially in their tail tendons. They found that estrogen treatment increased the stiffness and the melting temperature of collagenous tissues. While estrogen increased the cross-linking with aging, removing the pituitary gland was found to retard the aging. Later, the cross-linking enzymes transglutaminase and lysyl oxidase, which are induced by estrogen, were found to be a major factor in the cross-linking of collagen and other molecules. When estrogen was found to age the connective tissues, it was assumed that continual breeding during an animal's life-time, greatly increasing the total exposure of the tissues to estrogen, would increase the aged rigidity of the connective tissues, but these animals were found to have less rigid tissues. During pregnancy other hormones, especially progesterone, were also increased, and it was suggested that this reversed the effects of aging and estrogen. Since most people had believed that frequent pregnancies would cause a woman to age more rapidly, a large survey of records was done, to compare the longevity of women with the number of pregnancies. It was found, in the very extensive Hungarian records, that life-span was increased in proportion to the number of pregnancies. Despite these very interesting results in the 1950s and 1960s, the growing influence of the estrogen industry changed the direction of aging research, favoring the belief that decreasing estrogen accelerated the deterioration of tissues in aging, and the popularity of Denham Harman's "free radical theory of aging" led many people to assume that random reactions produced by lipid peroxidation were responsible for most of the cross-linking, and that theory was gradually replaced by the "glycation" theory of aging, in which sugar molecules break down and form the cross-links, by random, non-enzymic processes. Estrogen's role in aging was completely by-passed. The meat industry is interested in reducing the toughness of meat, by influencing the nature of the collagen in muscle. Castrated animals were found to produce meat that was tenderer than that of intact males. When castrated animals were treated with testosterone, the amount of collagen was increased, making the meat tougher. But when dihydrotestosterone, which can't be converted to estrogen was used, the meat didn't become tough. Treatment with estrogen produced the same increase of collagen as treatment with testosterone, showing that testosterone's effect was mainly the result of its conversion to estrogen (Miller, et al., 1990). In the 1960s and '70s the estrogen industry was looking for ways to build on the knowledge that in puberty estrogen is responsible for accelerating the calcification of the growth plate at the ends of the long bones, and to find a rationale for selling estrogen to all women concerned with the problems of aging. As bone metabolism was investigated, two kinds of cell were found to be active in constantly remodeling the bone structure: Osteoclasts (breaking it down), and osteoblasts (building new bone). Estrogen was found to slow the actions of the osteoclasts, so the idea that it would delay osteoporosis became the basis for a huge new marketing campaign. Slowing bone metabolism became the focus. Although estrogen was known to increase prolactin, and prolactin was known to accelerate bone loss, nearly all publications began to focus on substances in the blood or urine that corresponded to the rate of bone turnover, with the implication that increasing bone turnover would correspond to a net loss of bone. This was the context in which, during the 1980s, articles about thyroxine's role in causing osteoporosis began to appear. The thyroid hormone supports bone renewal, and increases indicators of bone breakdown in the blood and urine. If estrogen's use was to be justified by slowing bone turnover, then the effects of thyroid, accelerating bone turnover, should be interpreted as evidence of bone destruction. A basic problem with many of the publications on thyroid and bone loss was that they were talking about an unphysiological medical practice (prescribing the pre-hormone, thyroxine), which frequently failed to improve thyroid function, and could even make it worse, by lowering the amount of T3 in the tissues. Later, it was noticed that high TSH was associated with the signs of lower bone turnover. TSH rises when there is less thyroid hormone, but (after the recombinant TSH became available for medical use) a few publications argued that it was the TSH itself, rather than the absence of thyroid hormone, that was "protecting" the bones

(lowering the evidence of bone turnover). The doctrine that had been developed to support estrogen therapy was now used to oppose thyroid therapy. Keeping the TSH high would slow bone turnover. Working in this cultural context, genetic engineers at Amgen identified a protein that inhibited the formation of osteoclast cells, and slowed bone metabolism. It was suggested that it was responsible for estrogen's suppression of the osteoclasts, and many publications appeared showing that it was increased by estrogen. It was named "osteoprotegerin," meaning "the bone protecting protein." Prolactin increases osteoprotegerin (OPG), reducing bone resorption just as estrogen does. Serotonin also increases OPG, and it turns out that OPG is elevated in all of the pathological conditions associated with high serotonin, including cancer, pulmonary artery hypertension, vascular calcification, and even bone loss. While Arthur Everitt, Verzar, and others were studying the effects of the rat's pituitary (and other glands) on collagen, W. D. Denckla investigated the effects of reproductive hormones and pituitary removal in a wide variety of animals, including fish and mollusks. He had noticed that reproduction in various species (e.g., salmon) was quickly followed by rapid aging and death. Removing the pituitary gland (or its equivalent) and providing thyroid hormone, he found that animals lacking the pituitary lived much longer than intact animals, and maintained a high metabolic rate. Making extracts of pituitary glands, he found a fraction (closely related to prolactin and growth hormone) that suppressed tissue oxygen consumption, and accelerated the degenerative changes of aging. Aging, estrogen, cortisol, and a variety of stresses, including radiation and lipid peroxidation, chemically alter collagen, producing cross-links that increase its rigidity, and affect the way it binds minerals. The cross-linking enzymes induced by estrogen are involved in the normal maturation of bone collagen, and at puberty when estrogen increases, bone growth is slowed, as the cross-linking and mineralization are accelerated. With aging and the accumulation of heavy metals and polyunsaturated fats, random oxidative processes increase the cross-linking. In bones, the relatively large masses of cartilage absorb oxygen and nutrients slowly, so internally the amount of oxygen is very limited, about 1/5 as much as at the surface, and this low oxygen tension is an important factor in regulating growth, differentiation, cross-linking, and calcification, maintaining bone integrity. But in blood vessels the connective tissues are abundantly supplied with oxygen and nutrients; this is normally a factor regulating the production of collagen and its cross-linking, and preventing calcification. When the factors promoting collagen synthesis and maturation are increased systemically, with aging and stress, the excess cross-linking slows the biological renewal process in bones, but in blood vessels the same processes creating excess cross-linking initiate a calcification process, involving the various factors that in youth are responsible for normal maturation of bone. Prolactin, like estrogen, interferes with thyroid function and oxygen consumption (Wade, et al., 1986; Strizhkov, 1991; Spatling, et al., 1982). Many years ago, repeated lactation was considered to cause osteoporosis and loss of teeth, and prolactin, which mobilizes calcium from bones for the production of milk, was recognized as an important factor in bone loss. Drugs that increase prolactin were found to cause osteoporosis. In the 40 years since the drug industry began its intense promotion of estrogen to prevent and treat osteoporosis, there has been very little attention to the fact that estrogen increases prolactin, which contributes to osteoporosis, but some people (e.g., Horner, 2009) have noticed that oral contraceptives and menopausal hormone treatments have damaged the bones of the inner ear, causing otosclerosis and impaired hearing, and have suggested that prolactin mediates the effect. A few years ago, the "serotonin reuptake inhibitor" antidepressants, already known to increase prolactin by increasing the effects of serotonin, were found to be causing osteoporosis after prolonged use. Estrogen increases serotonin, which besides promoting the secretion of prolactin, also stimulates the production of parathyroid hormone and cortisol, both of which remove calcium from bone, and contribute to the calcification of blood vessels. The association between weakened bones and hardened arteries is now widely recognized, but researchers are being careful to avoid investigating any mechanisms that could affect sales of important drug products, especially estrogen and antidepressants. Following the recognition that the SSRI drugs were causing osteoporosis, it was discovered that the serotonin produced in the intestine causes bone loss, and that inhibiting intestinal serotonin synthesis would stop bone loss and produce a bone building anabolic effect (Inose, et al., 2011). One group that had been concentrating on the interactions of genes commented that, recognizing the effects of intestinal serotonin, they had suddenly become aware of "whole organism physiology" (Karsenty and Gershon, 2011). In previous newsletters I have talked about the ability of intestinal irritation and the associated increase of serotonin to cause headaches, asthma, coughing, heart and blood vessel

disease, muscular dystrophy, flu-like symptoms, arthritis, inflammation of muscles and nerves, depression, and inflammatory brain diseases. With the new recognition that serotonin is a basic cause of osteoporosis, intestinal health becomes a major issue in aging research. The protein that inhibits intestinal formation of serotonin is the low density lipoprotein receptor-related protein. This seems likely to have something to do with the fact that "low" HDL is associated with better bones. A low level of LDL is associated with increased vertebral fractures (Kaji, et al., 2010). Cartilage synthesis and turnover are highest at night. It is inhibited by metabolic acidosis (increased lactic acid), but not by respiratory acidosis (CO₂) (Bushinsky, 1995). Since most calcium is lost from bone during the night (Eastell, et al., 1992; even in children: DeSanto, et al., 1988) in association with the nocturnal rise of the catabolic substances, such as free fatty acids, cortisol, prolactin, PTH, and adrenalin, things which minimize the nocturnal stress can decrease the bone turnover. These include calcium (Blumsohn, et al., 1994) and sugar. Catabolic substances and processes increase with aging, especially at night. Babies grow most during the night when bone turnover is high, and even a daytime nap accelerates collagen turnover (Lutchman, et al., 1998). Discussions about whether a certain person's osteoporosis is "menopausal osteoporosis" or "senile osteoporosis" have neglected the possibility that osteoporosis doesn't begin in either menopause or old age, but that it is the result of life-long developmental processes that interact with all the factors that are involved in aging. The fact that the collagen content of old bone is lower than in young bone (as a percentage of bone weight) shows that the problem in osteoporosis isn't a lack of calcification, it's a deficiency of tissue renewal, parallel to sarcopenia, the decrease of muscle mass with aging. Systemically decreased tissue renewal would account for the association of bone loss with other processes such as male baldness (Morton, et al., 2007) and Alzheimer's disease (Zhou, et al., 2011, Duthie, et al., 2011). A high level of respiratory energy production that characterizes young life is needed for tissue renewal. The accumulation of factors that impair mitochondrial respiration leads to increasing production of stress factors, that are needed for survival when the organism isn't able to simply produce energetic new tissue as needed. Continually resorting to these substances progressively reshapes the organism, but the investment in short-term survival, without eliminating the problematic factors, tends to exacerbate the basic energy problem. This seems to be the reason that Denckla's animals, deprived of their pituitary glands, but provided with thyroid hormone, lived so long: they weren't able to mobilize the multiple defenses that reduce the mitochondria's respiratory energy production. Several things that the geneticists would never be able to fit into their schemes of "bone regulatory molecules" such as OPG, growth hormone, parathyroid hormone, and estrogen, fit neatly with the idea that bone health is maintained by respiratory energy and tissue renewal, under the influence of thyroid hormone. For example, adrenaline, which is increased by stress, aging, and hypothyroidism (and in many cases by estrogen), causes bone loss. Even the bone loss caused by immobility can be blocked by an adrenaline blocker such as propranolol. (The stress of immobility also famously increases serotonin.) Adrenaline tends to decrease carbon dioxide and increase lactic acid, and it strongly increases parathyroid hormone (Ljunhggall S, et al., 1984). Calcium activates mitochondrial respiration, and lowers adrenaline (Luft, et al., 1988), parathyroid hormone (Ohgitani, et al., 1997), and prolactin (Kruse and Kracht, 1981). Copper, which is the co-factor for the cytochrome C oxidase enzyme, activated by thyroid, is essential for bone formation and maintenance, and is consistently deficient in osteoporosis. Thyroid hormone increases the body's ability to assimilate copper. Aspirin, which stimulates bone formation, has other thyroid-like actions, including activation of mitochondrial respiration and energy production, with an increase of cytochrome C oxidase (Cai, et al., 1996), and it lowers serotonin (Shen, et al., 2011). It also apparently protects against calcification of the soft tissues, (Vasudev, et al., 2000), though there has been surprisingly little investigation of that. "Aspirin can promote trabecular bone remodeling, improve three-dimensional structure of trabecular bone and increase bone density of cancellous in osteoporotic rats by stimulating bone formation. It may become a new drug for the treatment of osteoporosis" (Chen, et al., 2011). A wide range of inflammatory mediators that accelerate inflammation and bone loss also inhibit thyroid function. People who ate more polyunsaturated fat, which inhibits thyroid and oxidative metabolism, were several times more likely to have osteoporotic fractures (that is, essentially spontaneous fractures) than people who ate the least (Martinez-Ramirez, et al., 2007). Arachidonic acid stimulates prolactin secretion, and prolactin acts on the thyroid gland to decrease its activity, and on other tissues to increase their glycolysis (with lactate production), while decreasing oxidative metabolism (Spatling, et al., 1982; Strizhkov, 1991). Living at high altitude, which strengthens bones, increases thyroid activity and decreases

prolactin (Richalet, et al., 2010) and parathyroid hormone (Khan, et al., 1996). It lowers free fatty acids, which lower bone mass by reducing bone formation and increasing bone resorption (Chen, et al., 2010). In menopausal women, polyunsaturated fatty acids and even monounsaturated fats are associated with bone loss, fruit and vegetable consumption protects against bone loss (Macdonald, et al., 2004). While it's very interesting that the drug propranolol which blocks adrenaline, and drugs that block serotonin formation, have bone protective and restorative effects, they also have undesirable side effects. Food choices that optimize oxidative metabolism are the safest, as well as the most economical, way to approach the problem of osteoporosis and other degenerative changes. A person can easily perceive changes in appetite, quality of sleep, changes in skin, hair, and mood, etc., but blood tests could be used to confirm that the right choices were being made. Tests for vitamin D, parathyroid hormone, free fatty acids, and CO₂/bicarbonate, as well as the hormones, can be helpful, if a person isn't sure whether their diet, sunlight exposure, and thyroid supplementation is adequate. The popular medical understanding of the organism is based on a mechanistic view of causality, in which genes have a central role, causing things to develop and function in certain ways, and that hormones and drugs can cause genes to increase or decrease their activity. Genes that build bones can be activated by one substance, and genes that tear down bones can be inhibited by another substance. The "osteoprotegerin" story illustrates the problem with that kind of thinking. Vernadsky's description of an organism as a "whirlwind of atoms" is probably a better way to think of how "causality" works. The moving air in a whirlwind forms a self-intensifying system, with the motion reducing the pressure, causing more air to be drawn into the system. The atoms moving in coordination aren't acting as separate things, but as parts in a larger thing. The way in which increased metabolism in the bones acts favorably on the metabolism of kidneys, blood vessels, lungs, liver, digestive system, etc., which in turn favors the bones' renewal, is analogous to the tendency of a whirlwind to intensify as long as there is a source of energy. The intensity of oxidative metabolism is the basic factor that permits continuing coordination of activity, and the harmonious renewal of all the components of the organism.

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